

T3 group factors: EFT_1_after_F

Active theory: $\text{SM} + S_1 + S_2 + C_{LLS_1S_2}$. The fermion F is absent, so y_1, y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1, m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2} \lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HS_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$(S_1^\dagger S_2)(S_2^\dagger S_1)$

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-E	$(3, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	F	S1, S2

Field representations

run	S_1	S_2	α
T3 - E	$(3, -2)$	$(3, -1)$	-4

Gauge and Yukawa beta functions in this EFT

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	T3 - E ($\alpha = -4$)
g_1^3	$\frac{71}{6}$

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
g_2^3	$-\frac{11}{6}$

 $\beta_{g_3}^{(1)}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
g_3^3	-7

 $\beta_{Y_u}^{(1)}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

 $\beta_{Y_d}^{(1)}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

 $\beta_{Y_e}^{(1)}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

 $\beta_{m_1^2}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$\lambda_{S_1}^{(1)} m_1^2$	8
$g_2^2 m_1^2$	-12
$g_1^2 m_1^2$	-24
$\lambda_{12}^{(1)} m_2^2$	6
$\lambda_{S_1}^{(A)} m_1^2$	4

$\lambda_{12}^{(\times)} m_2^2$	4
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$\beta_{m_2^2}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$\lambda_{S_2}^{(1)} m_2^2$	8
$g_2^2 m_2^2$	-12
$g_1^2 m_2^2$	-6
$\lambda_{12}^{(1)} m_1^2$	6
$\lambda_{S_2}^{(A)} m_2^2$	4
$\lambda_{12}^{(\times)} m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$ \lambda_5 ^2$	4
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{51}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	6
g_2^4	6
g_1^4	12
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	4
$\lambda_{HS_1}^{(1)} T$	2

$\beta_{\lambda_{HS_2}^{(1)}}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$ \lambda_5 ^2$	4
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	6
g_2^4	6
g_1^4	3
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	4
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	4
$\lambda_{HS_2}^{(1)} T$	2

$\beta_{\lambda_{12}^{(1)}}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$ \lambda_5 ^2$	$\frac{4}{3}$
$g_2^2 \lambda_{12}^{(1)}$	-24
$g_1^2 \lambda_{12}^{(1)}$	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4
g_2^4	12
g_1^4	48
$\left(\lambda_{12}^{(A)}\right)^2$	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	4
$\left(\lambda_{12}^{(\times)}\right)^2$	4
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	4

β_{λ_5}

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$\lambda_5 \lambda_1$	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4
$\lambda_5 \lambda_{12}^{(1)}$	2
$g_2^2 \lambda_5$	$-\frac{33}{2}$
$g_1^2 \lambda_5$	$-\frac{33}{2}$
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2
$\lambda_5 \lambda_{HS_2}^{(A)}$	2
$\lambda_5 \lambda_{12}^{(A)}$	2
$\lambda_5 \lambda_{12}^{(\times)}$	-2
$\lambda_5 T$	2

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$g_1^2 g_2^2$	-24

$\beta_{\lambda_{HS_2}^{(A)}}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$g_1^2 g_2^2$	-12

$\beta_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
$g_1^2 g_2^2$	48

$\beta_{\lambda_{12}^{(\times)}}$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
g_2^4	6
$\left(\lambda_{12}^{(A)}\right)^2$	2
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	-2
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	-2

Full Weinberg-coefficient RGE in EFT1

Writing the canonical mixed LLSS coefficient as C_{12} , the coupled one-loop Weinberg equation used in the F -first EFT is

$$16\pi^2 \frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \quad (1)$$

$$- \frac{3}{2} [(Y_e^\dagger Y_e)^T C_5 + C_5 (Y_e^\dagger Y_e)] \quad (2)$$

$$+ R_W \lambda_5 C_{12}. \quad (3)$$

Note: $\frac{7}{2}$ thing isnt there if we integrate out our fermion F so its not here

Direct $C_{12} \rightarrow C_5$ recoupling factor

For each representation the source term is

$$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{12}.$$

$\mathcal{S}$	T3 – E ($\alpha = -4$)
R_W	$-\frac{4\sqrt{6}}{3}$

T3 – E ($\alpha = -4$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{3} y_1 y_2}{3 M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{6} y_1 y_2}{3 M_F}$	$\mathcal{I}_{LLS_1 S_2}$

Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched C_5 boundary condition and the subsequent SMEFT running is universal.